

CS3300 - Compiler Design

Introduction

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IIT Madras

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Administrivia

- Lecture Timings
 - Slot B: Monday 9 AM, Tuesday 8 AM, Wednesday 1 PM, Friday 11 AM
- Lab Timings
 - Slot Q: Tuesday 2 PM to 4:45 PM
- Course Webpage:
https://fplaunchpad.org/cs3300_m26/
 - Lecture slides, assignment deadlines and lab session dates.
 - No lab in the first week. The first lab session is the Flex and Bison tutorial.
- Course Moodle page: TBD
- Slack for communication: TBD
- Instructor e-mail address: kcsrkc@cse.iitm.ac.in
 - Instructor Office Hours: None.
 - Feel free to ping me on Slack if you want to meet.

Grading Policy

- Theory: 60%, Lab: 40%
- Theory
 - 3 short quizzes: 3% each
 - 15 minutes, in class, multiple choice or one-line answers.
 - Roughly midway between the start and Quiz 1, between Quiz 1 and Quiz 2, and between Quiz 2 and the endsem.
 - Quiz 1: 12%, Quiz 2: 12%, Endsem: 27%
 - **Extra:** Class Participation: upto 5%.
 - Class Participation will be monitored throughout the semester. You can participate by asking/answering questions during the lectures and/or in the Slack.
- Lab: 6 assignments, 15 marks each (90 marks in total)
 - **9 marks** take-home, **6 marks** for an extension written in the lab.

Assignments: take-home, then lab

- Six assignments. You build a compiler for a subset of Java, one phase at a time.
- Each is marked out of 15: **9** for the take-home part, **6** for an extension you write during the lab session after the deadline.
- In the lab you extend **the code you submitted**, on DCF machines, with no network access other than Moodle.
 - The extension is not announced in advance.
 - It is small. The question is whether you can work in your own code.
- Lab attendance is mandatory. Miss a session without an institute-approved reason and your take-home marks for that assignment are capped at 20%.
- If your own submission will not build, ask for the reference implementation and extend that instead; that assignment is then capped at 50%.

Why it works this way

- These assignments can be produced in one shot by an LLM, and solutions to them circulate.
- So the take-home is where you learn the phase. The lab is where you show that you did.
- So do the take-home yourself. Code you did not write will cost you the lab marks: 6 per assignment, 16% of your final grade.

LLM policy

- **No LLM use is permitted in this course**, for any part of any assignment or evaluation.
- Using one is treated as **plagiarism**, and the plagiarism policy is enforced strictly.
- The point is for you to learn to build a compiler. Code you did not write teaches you nothing.

If you are caught, the Institute's graded punishments apply:

- First instance: **zero** for that assignment, **one grade lower** in the course, and 50 hours of library work.
- Repeated: **U grade** in the course.
- Either way, you become ineligible for a supplementary exam or contact course in this subject.

Course outline

- Overview of Compilers
- Lexical Analysis and Parsing
- Type checking
- Intermediate Code Generation
- Register Allocation
- Code Generation
- Overview of advanced topics.

Goal of the course: At the end of the course,

- Students will have a fair understanding of some standard passes in a general purpose compiler.
- Students will have hands on experience on implementing a compiler for a subset of Java.

- Compilers: Principles, Techniques, and Tools, Alfred Aho, Monica Lam, Ravi Sethi, Jeffrey D. Ullman. Addison-Wesley, 2007 [**The Dragon Book**].
 - **2020 ACM Turing Award!** – . . . Who Shaped the Foundations of Programming Language Compilers and Algorithms
 - *“Columbia’s Aho and Stanford’s Ullman Developed Tools and Fundamental Textbooks Used by Millions of Software Programmers around the World”*
- Modern compiler implementation in Java, Second Edition, Andrew W. Appel, Jens Palsberg. Cambridge University Press, 2002.

Your friends: Languages and Tools

Start exploring

- C and Java – familiarity is a must. Use an editor or IDE you are comfortable with; VS Code is a reasonable default.
- Flex, Bison, JavaCC, JTB – tools you will learn to use.
- Make and shell scripts – you will want them for running your compiler over test cases.

Mutual expectations

For the class to be a mutually learning experience:

- What will be required from the students?
 - An open mind to learn.
 - Curiosity to know the basics.
 - Explore their own thought process.
 - Help each other to learn and appreciate the concepts.
 - Honesty and hard work.
 - Leave the fear of marks/grades.
- What are the students expectations?

Acknowledgement

These slides are heavily adapted from the slides prepared by V Krishna Nandivada and Kartik Nagar @ IIT Madras. Liberal portions of text are also taken verbatim from Antony L. Hosking @ Purdue, Jens Palsberg @ UCLA, Alex Aiken @ MIT and the Dragon book.

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What, When and Why of Compilers

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Compilers: What?

- What is a compiler?
 - a program that translates an executable program in one language into an executable program in another language.
 - Usually from a high-level language to machine language
- What is an interpreter?
 - a program that reads an executable program and produces the results of running that program
 - usually, this involves executing the source program in some fashion
- This course deals mainly with compilers. Many of the same issues also arise in interpreters.
- A common statement – XYZ is an interpreted (or compiled) language.

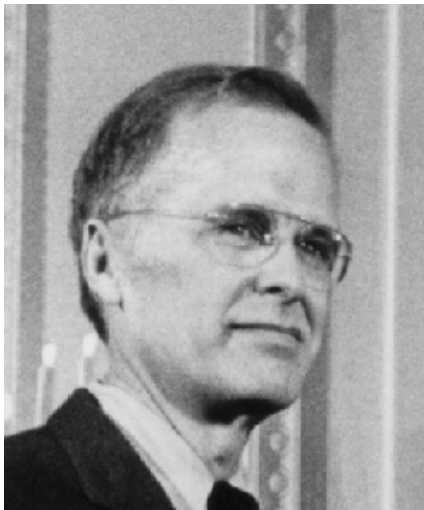
Examples

- “Low level” languages are typically compiled.
 - C, C++, Go, Rust
- “High level” languages are typically interpreted.
 - Python, Ruby
- Some languages are both compiled and interpreted
 - Java, Javascript - Interpreter + Just in Time (JIT) Compiler

Compilers: When?

- In 1954, IBM developed the 704, “the first mass-produced computer with floating-point arithmetic hardware” [Wikipedia].
 - Unfortunately, software costs would exceed hardware costs, since all programming was done in assembly.
- **John Backus** developed the FORTRAN I language (1957) for writing high-level code, and also a compiler for translating it to assembly.
 - Development time halved, with performance being close to the hand-written assembly!
 - Modern compilers preserve the outline of the FORTRAN I compiler
- Independently, in the 1950s, **Grace Hopper** developed the COBOL language and a compiler for it.

Images of the day



Turing Award Winners, Grace Hopper and John Backus

Compilers: Why?

Isn't it a solved problem? “Optimization for scalar machines was solved years ago”

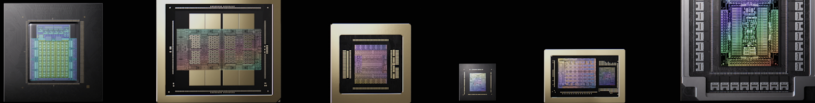
Machines have changed drastically in the last 20 years
Changes in architecture $\Rightarrow$ changes in compilers

Major unsolved problems: Design of a programming language and its compiler for

- AI/ML
- many- and multi-core architectures
- intermittent computing
- FPGAs
- probabilistic programming languages

AI/ML Hardware Innovation requires . . .

NVIDIA Rubin Platform
Six new chips. One AI supercomputer.



The image displays six distinct NVIDIA Rubin Platform chips arranged horizontally. From left to right: 1. Vera CPU, a square chip with a grid of green and blue cores. 2. Rubin GPU, a larger square chip with a complex, multi-colored circuit pattern. 3. NVLink 6 Switch, a square chip with a purple and blue circuit pattern. 4. ConnectX-9, a small square chip with a blue circuit pattern. 5. BlueField-4 DPU, a rectangular chip with a purple and blue circuit pattern. 6. Spectrum-6 Ethernet Switch, a larger rectangular chip with a complex, multi-colored circuit pattern and a metal frame.

Vera CPU 88 Olympus Cores LPDDR5	Rubin GPU 50PF HBM4	NVLink 6 Switch Single Hop 3,600 GB/s	ConnectX-9 800 Gb/s per port	BlueField-4 DPU 64C Grace CPU + CX-9 800 Gb/s	Spectrum-6 Ethernet Switch 102.4 Tb/s, CPO
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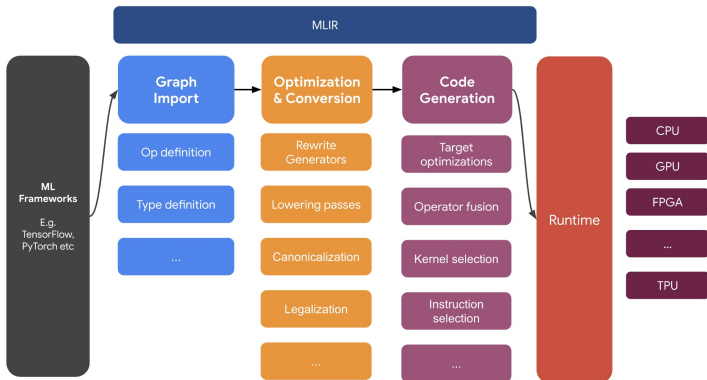
Six chips in one platform. Every one of them is a compilation target.

... Compiler Innovation

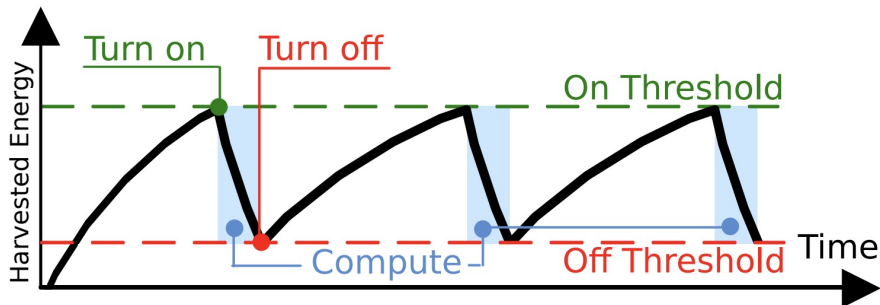
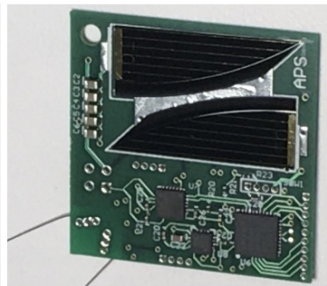
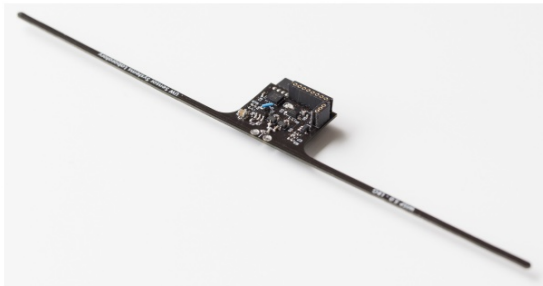
- None of that silicon is usable without a compiler. CUDA (2007) is what turned a graphics chip into a general purpose parallel machine.
- `nvcc` takes one source file and splits it across host and device; the libraries on top of it are generated and tuned per architecture.
- Other vendors have built comparable silicon. What is harder to copy is two decades of compiler and library work.
- We call NVIDIA a hardware company. A great deal of what you are paying for is the toolchain.

Compiler Innovation

MLIR, or Multi-Level Intermediate Representation, sits between the model representation and low-level compilers/executors that generate hardware-specific code.



Intermittent computing



Compiler construction is a microcosm of computer science

- **Algo** graph algorithms, union-find, dynamic programming, ...
- **theory** DFAs for scanning, parser generators, lattice theory, ...
- **systems** allocation, locality, layout, synchronization, ...
- **architecture** pipeline management, hierarchy management, instruction set use, ...
- **optimizations** Operational research, load balancing, scheduling, ...

Inside a compiler, all these and many more come together. Has probably the healthiest mix of theory and practise.

Compiler Design is challenging and fun

- interesting problems
- primary responsibility (read:*blame*) for performance
- new architectures $\Rightarrow$ new challenges
- *real* results
- extremely complex interactions

Compilers have a major impact on how computers are used

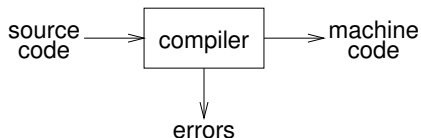
Overview of Compilers

Requirements

What qualities are important in a compiler?

- 1 Correct code
- 2 Output runs fast
- 3 Compiler runs fast
- 4 Compile time proportional to program size
- 5 Good diagnostics for syntax errors
- 6 Works well with the debugger
- 7 Consistent, predictable optimization

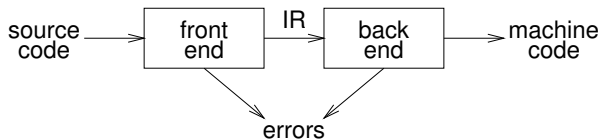
Abstract view



Implications:

- recognize legal (and illegal) programs
- generate correct code
- manage storage of all variables and code
- agreement on format for object (or assembly) code

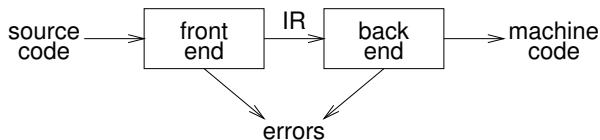
Traditional two pass compiler



Implications:

- intermediate representation (IR).
- front end maps legal code into IR
- back end maps IR onto target machine
- simplify retargeting
- allows multiple front ends
- multiple passes $\Rightarrow$ better code

Traditional two pass compiler

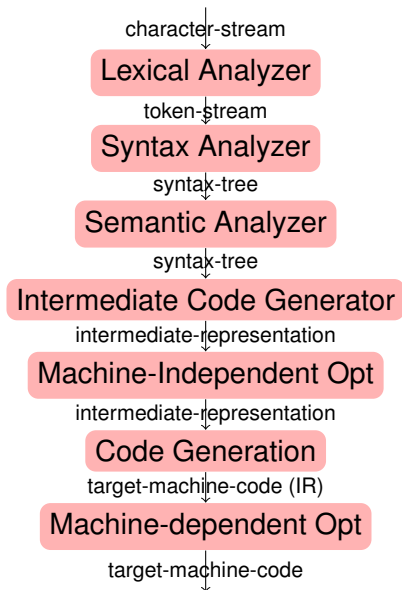


A rough statement: Most of the problems in the Front-end are simpler (polynomial time solution exists).

Most of the problems in the Back-end are harder (many problems are NP-complete in nature).

Our focus: Mainly front end and little bit of back end.

Phases inside the compiler



Front end responsibilities:

- Recognize syntactically legal code; report errors.
- Recognize semantically legal code; report errors.
- Produce IR.

Back end responsibilities:

- Optimizations, code generation.

Our target

- five out of seven phases.
- glance over optimizations

Major Phases in a Compiler

- Lexical Analysis
- Parsing
- Semantic Analysis
- Optimization
- Code Generation

Lexical Analysis

- To understand the different phases of a compiler, we will use as an analogy, how humans comprehend the English language.
- First step: recognize words in a sentence
 - Smallest unit above letters.
- All words in a sentence must be valid. Examples:
 - This is a sentence
 - Thsi si otn

Lexical Analysis for Programs

- Lexical Analyzer divides program text into 'words' or 'tokens' (in compiler terminology)
- For example, consider the program statement:
`if x==y then z=1; else z=2;`
- Tokens
 - Keywords: `if, then, else`
 - Identifiers: `x, y, z`
 - Operators: `==, =, ;`
 - Constants: `1, 2`
 - Whitespace
- Lexical Analyzer also rejects a program if it has any invalid word.

Parsing

- Once words are understood, the next step is to understand the structure of the sentence.
- Parsing = Diagramming Sentences
 - The diagram will be a tree.

This is a sentence

Parsing Programs

- Parsing program statements is the same.

```
if x==y then z=1; else z=2;
```

- Once sentence structure is understood, we can try to understand its “meaning”.
 - Understanding the entire meaning of a program is too hard for compilers—in fact, undecidable.
- Compilers perform limited semantic analysis to catch inconsistencies.

Semantic Analysis in English

- Example: Ram said Shyam left his assignment at home.
 - What does “his” refer to? Ram or Shyam?
- Even worse: Ram said Ram left his assignment at home
 - How many Ram's are there? Which one left his assignment?

Semantic Analysis in Programming

- Programming languages define strict rules to avoid such ambiguities.
- The C code prints “4”; the inner definition is used

```
{  
    int X = 3;  
    {  
        int X = 4;  
        printf("\%d", X);  
    }  
}
```

More Semantic Analysis

- Compilers perform many semantic checks besides variable bindings.
- Example: **Shyam left her homework at home**
- Possible type mismatch between **her** and **Shyam**

Optimization

- No strong counterpart in terms of our analogy, but we can think of it as editing to make sentences simpler.
- Automatically modify programs so that they
 - Run faster
 - Use less memory
 - In general, use of conserve some resource
- Example: $X = Y * 0$ is the same as $X=0$, but does not involve an additional multiplication, and is thus faster.
- In this course, we won't study optimization in detail. But there are advanced courses such as Modern Compilers: Theory and Practice for that purpose.

- Typically produces Assembly code.
- In terms of our analogy, translating the sentence in English into another language.

Intermediate Representations

- Compilers typically perform translations between successive intermediate languages.
 - All but the first and last are intermediate representations (IR) internal to the compiler.
- IRs are ordered in descending level of abstraction
 - Highest is source.
 - Lowest is assembly.
 - Other IRs that we will during the course: Token Stream, Syntax Tree, Three-Address Code.
- IRs are useful because lower levels expose features hidden by higher levels
 - Registers, Memory layout, raw pointers, etc.
- But lower levels obscure high-level meaning
 - Classes, Higher-order functions, Even loops...